

Geography Ch-10 (Advanced) — The Indian Coast: East vs West & Coastal Management

Advanced / India-applied | GS-I & GS-III | Source: D.R. Khullar (India: A Comprehensive Geography)

West coast (submergent) vs East coast (emergent, deltaic); backwaters, lagoons

Extra CA: CRZ 2019 & Blue Economy (distinct from Must-Do's NCCR erosion CA)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

ADVANCED — India's Coastal Plains: West vs East (Khullar)

■ NEWS TRIGGER

India's ~6,000-km coastal plains run from the Rann of Kachchh to the Ganga-Brahmaputra delta, split by the peninsula into a narrow submergent West Coast and a wide deltaic East Coast.

■ INTRO & ORIGIN

Khullar divides the coast into the **West Coastal Plain** (Western Ghats-Arabian Sea) and the **East Coastal Plain** (Eastern Ghats-Bay of Bengal), with contrasting form and origin.

*The West Coast is largely **submergent** (drowned, narrow, with estuaries and backwaters); the East Coast is **emergent** and built by **alluvial deltas** of the great east-flowing rivers.*

■ KEY DATA

Feature	West Coast	East Coast
Nature	Submergent, narrow, steep	Emergent, broad, gently sloping
Divisions	Konkan, Karnataka, Kerala (Malabar)	Northern Circars, Coromandel
Rivers/mouths	Estuaries (Narmada, Tapi)	Deltas (Mahanadi, Godavari, Krishna, Cauvery)
Water bodies	Backwaters/lagoons (Vembanad, Malabar)	Lagoons (Chilika, Pulicat)
Ports	Deep natural harbours (Mumbai, Marmagao)	Fewer natural harbours (mostly artificial)

■ STATIC THEORY

- The **West Coast** runs nearly straight N-S from Gujarat to Kanniyakumari, dotted with coves and creeks; the **Narmada and Tapi** form **estuaries** (not deltas).
- The **Malabar (Kerala) coast** has lakes, lagoons and **backwaters** — the largest being **Vembanad Lake**.
- The **East Coast** (Ganga delta to Kanniyakumari) is **deltaic** — built by the **Mahanadi, Godavari, Krishna and Cauvery**; it hosts **Chilika and Pulicat** lagoons.
- West-coast sub-divisions: **Konkan, Karnataka, Kerala** (Khullar); the East plains are largely **alluvial fillings** of the littoral zone.
- Contrast: the submergent West Coast gives deep natural harbours; the emergent, silt-laden East Coast has fewer natural harbours.

■ MUST-KNOW FACTS

1. West Coast = submergent, narrow (Konkan/Karnataka/Kerala).
2. Narmada & Tapi form **ESTUARIES** on the West Coast (not deltas).
3. Vembanad = largest Malabar backwater lake.
4. East Coast = emergent, deltaic (Mahanadi/Godavari/Krishna/Cauvery).

■ UPSC TRAPS

- ✗ Narmada and Tapi build deltas on the west coast.
- ✓ They form **ESTUARIES** (west-flowing through rift valleys); deltas are on the **EAST** coast.
- ✗ The East Coast is submergent with deep harbours.
- ✓ The East Coast is **EMERGENT** and deltaic, with fewer natural harbours; the **WEST** coast is submergent.

5. East-coast lagoons: Chilika, Pulicat.

✗ Vembanad is on the east coast.

✓ Vembanad is a Malabar (Kerala) WEST-coast backwater lake.

■ MAINS ANGLE

Contrast the origin and features of India's West and East coasts, and relate them to port geography, backwater ecology and deltaic livelihoods.

■ STUDY LINK

Geography → India → Coastal Plains (Khullar: West & East Coastal Plains)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — CRZ 2019 & the Blue Economy

GS-III · Indian coast applied → Coastal governance

■ NEWS TRIGGER

The Coastal Regulation Zone (CRZ) Notification 2019 governs construction along India's coast, while the Blue Economy vision seeks sustainable use of ocean resources — both shaped by erosion and sea-level risk.

■ INTRO & ORIGIN

CRZ zoning translates coastal geomorphology into land-use rules; the Blue Economy links that coast to fisheries, ports, tourism and marine energy.

■ STATIC THEORY

- **CRZ 2019:** No-Development Zones, Coastal Zone Management Plans (CZMPs) and **hazard lines** mapping sea-level-rise and erosion risk.
- **Blue Economy:** sustainable use of ocean resources — fisheries, aquaculture, shipping, tourism, marine renewable energy and biotech.
- Coastal erosion directly threatens the blue economy (fisheries, tourism, ports), so CRZ + mangrove shields aim to build resilience.

■ MUST-KNOW FACTS

1. CRZ 2019: NDZ + CZMP + hazard lines for coastal regulation.
2. Blue Economy = sustainable ocean-resource use (fisheries, ports, energy).
3. Erosion undermines the blue economy -> need integrated coastal management.

■ UPSC TRAPS

- ✗ CRZ rules apply uniformly to the whole coast.
- ✓ CRZ classifies the coast into categories (I-IV) with DIFFERENT rules for each.

■ MAINS ANGLE

Connect coastal geography to CRZ regulation and a resilient, sustainable blue economy.

■ STUDY LINK

Geography → Indian Coast → CRZ & Blue Economy (applied CA)